

Project Apex Race Report - Race Event

Executive Summary

The race analysis highlights varying performance across the Lamborghini field, with clear distinctions in raw pace, consistency, and operational efficiency. World Speed's Car #22 and Wayne Taylor Racing's Car #8 demonstrated exceptional performance, leading the field with average green flag paces of 2:09.552 and 2:09.620 respectively. Their remarkable consistency, with standard deviations of 0.261s and 0.432s, combined with highly efficient pit stops (1:36.673s and 1:36.735s average stationary times), underscore their dominant form. MLT Motorsports' Car #11 also showed strong pace (2:10.817) and good consistency (1.449s STDEV), positioning them firmly in the leading pack. Wayne Taylor Racing's Car #1, despite recording the fastest optimal lap (2:05.207) and fastest race lap (2:05.270), suffered from a significant driver performance delta (Hampus Eriksson was 17.513s slower than Danny Formal per best lap) and poor overall consistency (34.673s STDEV), preventing it from converting peak speed into sustained race leadership. The mid-field tier is characterized by mixed performance, often showing flashes of pace but struggling with consistency or operational inefficiencies. Alliance Racing's Car #28 (2:17.105 avg pace, 16.321s STDEV) and Precision Performance Motorsports' Car #46 (2:24.690 avg pace, 32.246s STDEV) exemplify this, both hampered by substantial driver pace differences (4.246s and 22.048s respectively). Flying Lizard Motorsports' Car #14, despite a good optimal lap (2:07.655) and minimal driver delta (0.093s), had an average green pace of 2:23.232 and consistency issues (26.327s STDEV), suggesting untapped potential. At the back of the field, RAFA Racing's Car #99 showed significant deficits in all areas, with the slowest average green pace (2:47.822) and worst consistency (37.927s STDEV). Their operational shortcomings were equally stark, recording the highest average pit cycle loss (324.5248s) and longest stationary times (2:57.817s). Flying Lizard Motorsports' Car #41 also struggled with pace (2:34.588), indicating a fundamental speed deficiency. The biggest strategic differentiator in this race was undoubtedly driver consistency, particularly the ability of all drivers within a car to maintain competitive lap times, closely followed by pit stop execution.

Tactical Insights

- {'team_type': 'leading', 'team': 'World Speed', 'car_number': '22', 'recommendation': 'Continue to refine tire management strategies and enhance data acquisition for deeper insights into tire degradation.', 'justification': 'Car #22 demonstrated exceptional race pace (2:09.552 average green) and driver consistency (0.261s

STDEV), coupled with efficient pit stops (1:36.673s average stationary time). While performing at a high level, the tire degradation model quality for Car #22 is rated "POOR" due to insufficient clean data (only 9 clean laps used for analysis). Improving the robustness of this data will enable more precise tire strategy calls and potentially unlock marginal gains, solidifying their competitive edge.'}

- {'team_type': 'mid-field', 'team': 'Wayne Taylor Racing', 'car_number': '1', 'recommendation': 'Implement a targeted driver development program for Hampus Eriksson to reduce the significant pace delta to teammate Danny Formal.', 'justification': 'Car #1 possesses the highest raw pace potential, evidenced by Danny Formal's optimal lap of 2:05.207. However, the team's average green pace (2:22.460) and race consistency (34.673s STDEV) are severely compromised by Hampus Eriksson being 17.513s slower per best lap time than Formal. This substantial performance gap is a primary limiter. Focused coaching, simulator work, and potentially tailored car setups for Eriksson are critical to bring his performance closer to his teammate's, allowing the car's full potential to be realized over a race distance.'}
- {'team_type': 'lagging', 'team': 'RAFA Racing', 'car_number': '99', 'recommendation': 'Urgently overhaul pit stop procedures and conduct a thorough review of driver performance and consistency.', 'justification': 'Car #99 exhibited the slowest average green pace (2:47.822) and the worst race pace consistency (37.927s STDEV). More critically, their pit stop performance was a significant handicap, recording the highest average pit cycle loss (324.5248s) and a stationary time of 2:57.817s across four stops. This indicates fundamental issues in pit lane operations. Immediate attention should be given to pit crew training, equipment efficiency, and strategic planning for pit windows. Concurrently, driver performance analysis and coaching are essential to address the severe on-track pace and consistency deficits.'}